States of matter, atoms and the Periodic Table

30 questions: 10 multiple choice, 20 structured.

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1 · Part 1

Paper 1 & 2 States of matter [1]

Which statement describes the particles in a solid?

A They are close together in a random arrangement and move past each other.

B They are close together in a regular arrangement and vibrate about fixed positions.

C They are far apart in a random arrangement and move quickly in all directions.

D They are close together in a regular arrangement and move past each other.

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2 · Part 1

Paper 1 & 2 States of matter [1]

At room temperature, solid carbon dioxide (dry ice) turns directly into carbon dioxide gas without becoming a liquid.

What is the name of this change of state?

A boiling

B condensing

C evaporating

D subliming

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3 · Part 1

Paper 2 only States of matter [1]

In an experiment, 14.8 g of ammonium chloride dissolves in 40.0 g of water at 20 °C to form a saturated solution.

What is the solubility of ammonium chloride at 20 °C, in g per 100 g of water?

A 37.0

B 14.8

C 27.0

D 5.92

4 · Part 1

Paper 1 & 2 States of matter [7]

The table gives the melting points and boiling points of four substances.

(a) Complete the table by giving the physical state of each substance at 25 °C.

Substance	Melting point in °C	Boiling point in °C	Physical state at 25 °C
oxygen	−218	−183	
bromine	−7	59	
mercury	−39	357	
sodium chloride	801	1413	

[2]

(b) Give the name of the substance in the table that is a liquid at −20 °C. [1]

(c) Bromine at 70 °C is cooled to 20 °C. State the name of the change of state that takes place. [1]

(d) Describe how the arrangement, the movement and the energy of the bromine particles change during this change of state. [3]

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5 · Part 1

Paper 1 & 2 | States of matter | [7]

Ammonia, NH_3, and hydrogen chloride, HCl, are both gases at room temperature. When they meet, they react to form ammonium chloride, a white solid.

In an experiment, a glass tube 100 cm long is clamped horizontally. At the same moment, a cotton wool plug soaked in ammonia solution is pushed into one end of the tube and a cotton wool plug soaked in hydrochloric acid is pushed into the other end. Ammonia gas is given off at one end and hydrogen chloride gas at the other. Both ends are then sealed with bungs.

After 8 minutes, a white ring of ammonium chloride forms inside the tube, 59 cm from the ammonia end.

(a) State the name of the process by which the gas particles spread along the tube. [1]

(b) Explain why the white ring forms closer to the hydrochloric acid end of the tube. [2]

(c) At room temperature, gas particles move at speeds of several hundred metres per second. Explain why it takes several minutes for the white ring to form. [2]

(d) The experiment is repeated with the tube at a higher temperature. Predict how the time taken for the white ring to form will change. Explain your answer. [2]

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6 · Part 1

Paper 1 & 2 | States of matter | [7]

Kofi places one small crystal of potassium manganate(VII) at the bottom of a beaker containing 100 cm^3 of water. He does not stir the water. The crystal dissolves, and after two hours all of the water is purple.

(a) State what is meant by the term solute. [1]

(b) Explain why the purple colour spreads through all of the water even though the water is not stirred. [2]

(c) Kofi labels the purple solution as solution 1. He transfers 10 cm^3 of solution 1 into 90 cm^3 of water and mixes it to make solution 2. He repeats this, each time using 10 cm^3 of the previous solution and 90 cm^3 of water, to make solutions 3, 4, 5 and 6. The table shows the colour of each solution. Calculate how many times more dilute solution 5 is than solution 1.

Solution	Colour
1	dark purple
2	purple
3	pale purple
4	very pale purple
5	faint pink
6	colourless

[1]

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(d) Explain what these results suggest about the particles in the crystal of potassium manganate(VII). [2]

(e) Kofi then adds potassium manganate(VII) crystals to a fresh sample of water until some solid remains undissolved. State what is meant by the term saturated solution. [1]

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7 · Part 1

Paper 2 only

States of matter

[7]

The solubility of potassium nitrate at 20 °C is 32 g per 100 g of water.

(a) State what is meant by the statement "the solubility of potassium nitrate at 20 °C is 32 g per 100 g of water". [2]

(b) Calculate the maximum mass of potassium nitrate that dissolves in 250 g of water at 20 °C. [1]

(c) Lena adds 50 g of potassium nitrate to 125 g of water at 20 °C. She stirs the mixture until no more solid dissolves. Determine the mass of potassium nitrate that does not dissolve. [2]

(d) Lena then warms the mixture to 50 °C. All of the remaining solid dissolves. Explain what this shows about the solubility of potassium nitrate. [2]

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8 · Part 1

Paper 2 only

States of matter

[8]

Ravi investigates the solubility of potassium chloride in water at 40 °C.

(a) Describe how Ravi should make a saturated solution of potassium chloride at 40 °C. [3]

(b) Ravi pours some of the saturated solution, with no undissolved solid, into a weighed evaporating basin and weighs it again. He then heats the basin until all of the water has evaporated, lets it cool and weighs it. The table shows his results. Calculate the solubility of potassium chloride at 40 °C in g per 100 g of water.

Measurement	Mass in g
empty evaporating basin	30.52
evaporating basin + saturated solution	54.43
evaporating basin + dry potassium chloride	37.38

[3]

(c) Ravi heats, cools and weighs the basin again, repeating this until the mass no longer changes. Explain why he does this. [2]

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9 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [1]

Which row correctly classifies chlorine, magnesium oxide and sea water?

	Element	Compound	Mixture
A	magnesium oxide	chlorine	sea water
B	chlorine	sea water	magnesium oxide
C	chlorine	magnesium oxide	sea water
D	sea water	magnesium oxide	chlorine

10 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [1]

Hexane (boiling point 69 °C) and heptane (boiling point 98 °C) are liquids that mix completely.

Which technique is used to separate a mixture of hexane and heptane into the two pure liquids?

A fractional distillation

B crystallisation

C filtration

D simple distillation

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11 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [7]

(a) The table describes the particles in four substances, P, Q, R and S. Complete the table by stating whether each substance is an element, a compound or a mixture.

Substance	Particles present	Element, compound or mixture
P	N_2 molecules only	
Q	CO_2 molecules only	
R	N_2 molecules and O_2 molecules	
S	Ne atoms only	

[4]

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(b) Hana mixes iron filings with powdered sulfur. Iron is attracted to a magnet but sulfur is not, so she can separate the iron from the mixture with a magnet. She then heats a fresh mixture of iron and sulfur strongly. A black solid called iron(II) sulfide, FeS, forms. Iron(II) sulfide is not attracted to a magnet. Explain, using these observations, the difference between a mixture and a compound. [3]

12 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [6]

(a) A sample of molten substance X is left to cool. Its temperature is recorded every minute. The table shows the results. Using the results, state the melting point of X.

Time in minutes	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Temperature in °C	88	80	73	69	69	69	69	69	64	58	53

[1]

(b) State the name of the change of state taking place between 3 minutes and 7 minutes. [1]

(c) Explain how the results show that X is a pure substance. [2]

(d) Predict how the results would be different if X contained a small amount of an impurity. [2]

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13 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [6]

Rock salt is a mixture of sodium chloride and sand. Sodium chloride is soluble in water. Sand is insoluble in water.

(a) Describe how you could obtain pure, dry crystals of sodium chloride from rock salt. [5]

(b) State the name given to the liquid that passes through the filter paper during filtration. [1]

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14 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [7]

Pure water can be obtained from sea water by simple distillation. Sea water is heated in a flask. The top of the flask is fitted with a thermometer, and a side arm leads to a water-cooled condenser. The liquid that leaves the condenser is collected in a beaker.

(a) Explain how simple distillation separates pure water from sea water. [4]

(b) The bulb of the thermometer is level with the side arm. State the temperature shown on the thermometer while pure water is being collected. [1]

(c) Explain why filtration cannot be used to obtain pure water from sea water, but simple distillation can. [2]

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15 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [7]

Mei is a food scientist. She uses paper chromatography, with water as the solvent, to find out which permitted food colourings are in an orange food colouring, X.

The table, from a reference book, gives the Rf values of five permitted food colourings when water is the solvent.

Food colouring	P	Q	R	S	T
Rf value in water	0.20	0.38	0.55	0.68	0.84

(a) On Mei's chromatogram, the solvent front moved 7.5 cm from the baseline. Colouring X separated into three spots. The table shows the distance each spot moved from the baseline. Calculate the Rf value of each spot.

Spot	Distance moved from baseline in cm
1	1.5
2	4.1
3	6.3

[3]

(b) Deduce which of the permitted food colourings P, Q, R, S and T are present in X. [2]

(c) Mei's colleague repeats the chromatography using ethanol as the solvent instead of water. Explain why the colleague cannot use the Rf values in the reference book to identify the

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colourings in X. [2]

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16 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [6]

Tomás investigates the dyes in a black ink using paper chromatography. This is his method.

- Draw a line in ink 1 cm from the bottom of a piece of chromatography paper.
- Put a small spot of the black ink on the line.
- Stand the paper in a beaker of water so that the water level is 2 cm from the bottom of the paper.
- Leave the paper until the water has nearly reached the top of the paper.

(a) Explain why the line should be drawn in pencil and not in ink. [2]

(b) Explain why the water level must be below the line. [2]

(c) Tomás corrects his method and repeats the experiment. The black ink produces three coloured spots above the line, and one spot that stays on the line. State what this result shows about the black ink. [1]

(d) Suggest why one spot stays on the line. [1]

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17 · Part 1 Paper 1 & 2 Atomic structure [1]

Which particle has a relative mass of 1 and a relative charge of +1?

A an electron

B a neutron

C an atom of hydrogen

D a proton

18 · Part 1 Paper 1 & 2 Atomic structure [1]

The table shows the numbers of sub-atomic particles in four atoms, W, X, Y and Z.

Atom	Number of protons	Number of neutrons	Number of electrons
W	6	6	6
X	7	7	7
Y	6	8	6
Z	5	6	5

Which two atoms are isotopes of the same element?

A W and Z

B W and Y

C X and Y

D X and Z

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19 · Part 1

Paper 1 & 2 Atomic structure [8]

(a) Use the Periodic Table to complete the table for each atom.

Atom	Number of protons	Number of neutrons	Number of electrons
fluorine-19			
calcium-40			
iron-56			

[3]

(b) State what is meant by the mass number of an atom. [1]

(c) Describe the structure of an atom. In your answer, give where each type of sub-atomic particle is found. [2]

(d) Explain why an atom has no overall electrical charge. [2]

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20 · Part 1

Paper 1 & 2 | Atomic structure | [7]

Naturally occurring magnesium is made of three isotopes: magnesium-24, magnesium-25 and magnesium-26.

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(a) Explain why magnesium-24 and magnesium-26 are isotopes of the same element. [2]

(b) The table shows the percentage abundance of each isotope in a sample of magnesium. Calculate the relative atomic mass (Ar) of magnesium in this sample. Give your answer to one decimal place.

Isotope	Percentage abundance
magnesium-24	79.0
magnesium-25	10.0
magnesium-26	11.0

[3]

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(c) Explain why the relative atomic mass of magnesium is not a whole number. [2]

21 · Part 1

Paper 1 & 2 Atomic structure [7]

Boron forms a compound called boron trifluoride, BF_3, which is made of molecules.

(a) State what is meant by the term molecule. [1]

(b) Boron has two isotopes, boron-10 and boron-11. State how an atom of boron-10 is different from an atom of boron-11. [1]

(c) In a sample of boron, the relative atomic mass is 10.8 to three significant figures. Without doing a calculation, explain how the value 10.8 shows that most of the atoms in the sample are boron-11. [2]

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(d) The sample contains only boron-10 and boron-11 atoms. Calculate the percentage of boron-10 atoms in the sample. [3]

22 · Part 1 Paper 1 & 2 The Periodic Table [1]

The elements in the Periodic Table are arranged in order of increasing value of one quantity.

What is this quantity?

A number of neutrons

B relative atomic mass

C atomic number

D mass number

23 · Part 1 Paper 1 & 2 The Periodic Table [1]

An element is in Group 6 and Period 3 of the Periodic Table.

What is the electronic configuration of an atom of this element?

A 2.8.6

B 2.6

C 2.8.3

D 2.8.8.6

24 · Part 1 Paper 1 & 2 The Periodic Table [1]

The positions of four elements in the Periodic Table are given.

Which element is a non-metal?

A the element in Group 1, Period 4

B the element in Group 2, Period 3

C the element in Group 3, Period 3

D the element in Group 7, Period 2

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25 · Part 1

Paper 1 & 2 The Periodic Table [6]

(a) Complete the table by giving the electronic configuration of an atom of each element.

Element	Atomic number	Electronic configuration
nitrogen	7	
aluminium	13	
chlorine	17	
calcium	20	

[4]

(b) Explain how the electronic configuration of chlorine shows its group and its period in the Periodic Table. [2]

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26 · Part 1

Paper 1 & 2 The Periodic Table [6]

The table gives the electronic configurations of atoms of five elements. The letters are not the chemical symbols of the elements.

Element	Electronic configuration
E	2.2
G	2.8.2
J	2.8.5
L	2.4
M	2.1

(a) Give the group and the period of element J. [2]

(b) Identify the two elements that are in the same group. Explain your answer. [2]

(c) State which two of the elements are non-metals. [1]

(d) Use the Periodic Table to state the name of element M. [1]

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27 · Part 1

Paper 1 & 2 The Periodic Table [6]

Three elements, X, Y and Z, are tested. The electrical conductivity of each solid element is measured. Each element is then burned in oxygen, and the oxide formed is shaken with water. The pH of the solution formed is measured. The table shows the results.

Element	Appearance	Electrical conductivity of the solid	pH of solution formed by the oxide
X	shiny grey solid	good	11
Y	yellow solid	poor	2
Z	black solid	good	5

(a) Classify elements X and Y as metals or non-metals. Give a reason for each answer using both sets of results. [2]

(b) Element Z is a non-metal. Explain why the results for Z could lead a student to classify Z wrongly, and state which result shows that Z is a non-metal. [2]

(c) Copper is a metal. Copper(II) oxide is a black solid that does not dissolve in water. Suggest a test, and its result, that would show that copper(II) oxide is a basic oxide. [2]

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28 · Part 1 Paper 1 & 2 The Periodic Table [6]

Lithium, sodium and potassium are in Group 1 of the Periodic Table. They all react with water in a similar way.

(a) Explain, in terms of electrons, why lithium, sodium and potassium have similar chemical properties. [2]

(b) Rubidium is below potassium in Group 1. Sodium reacts with water to form sodium hydroxide and hydrogen. Predict the names of the two products formed when rubidium reacts with water. [2]

(c) Silicon is in Period 3, the same period as sodium. Explain, using the electronic configurations of sodium and silicon, why silicon is not in the same group as sodium. [2]

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29 · Part 1

Paper 1 & 2 The Periodic Table [6]

Helium, neon and argon are noble gases in Group 0 of the Periodic Table.

(a) Give the electronic configurations of an atom of neon and an atom of argon. [2]

(b) Explain why the noble gases do not readily react. [2]

(c) The electronic configuration of helium is 2. Explain why helium is placed in Group 0, even though its atoms have only two electrons in the outer shell. [2]

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30 · Part 1 Paper 1 & 2 The Periodic Table [7]

Strontium has the atomic number 38.

(a) Use the Periodic Table to give the group and the period of strontium. [2]

(b) State the number of electrons in the outer shell of a strontium atom. [1]

(c) Predict whether solid strontium is a good or a poor conductor of electricity. Give a reason for your prediction. [2]

(d) Calcium oxide reacts with dilute hydrochloric acid to form calcium chloride and water. Predict the names of the products formed when strontium oxide reacts with dilute hydrochloric acid, and state what this reaction shows about the acid-base character of strontium oxide. [2]

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